Natural Capital and the Measurement of Productivity*

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Abstract

Current practice defines productivity as output that is not explained by measured economic inputs. Yet, many seemingly important inputs, e.g., natural capital, are omitted from the set of measured economic inputs and may confound measurement of productivity. We investigate how omitting natural capital as an input to production biases estimates of productivity growth. Empirical correlations between benchmark data on productivity growth and environmental change at the national level suggest the omission of natural capital biases these measures. We provide theoretical conditions under which better measurement of omitted environmental inputs reduces estimation error. Calibrated experiments based on data from the World Bank suggest that the inclusion of natural capital on the margin will improve measurement for a broad scope of countries.

Keywords: Aggregate Productivity; National Accounts; Natural Capital; Measurement

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*All errors are our own.

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Innovation and technological change over the past two centuries drove an unprecedented increase in economic activity. Industrialization simultaneously muddled societal awareness of fundamental connections between humans and nature. Society's interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital over time. Theodore Roosevelt noted this phenomenon at the outset of the 20th century:

> "With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature." (Roosevelt).

Formal systems of economic measurement have followed this trend; existing structures provide little guidance for separating nature's contribution to production from the contributions of human or produced capital. Human labor, vessels, and sonar are commonplace inputs to industrial fishing and measured by national accountants. Equally important is the fish stock and its ability to replenish over time. The fish stock, as an input to production, is often excluded from economic measurement. This continues to be the case despite managed fish stocks (those within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts (European Commission and Development).

This paper examines how the omissions of the environment as a factor in production affects a leading metric of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP is defined the growth in output unexplained by growth in inputs; it is has been a benchmark metric for economy-wide technological progress and innovation over the past century (Solow). Changes in TFP over time play a crucial role in shaping governments' budget projections and policy agendas over long time horizons (Congressional Budget Office; Council of Economic Advisers). Starting with Kuznet's seminal work report to congress in 1934, the broader development of statistical, economic, and environmental policy, has focused on measuring output and productivity growth by tracking capital and labor income over time (Kuznets). While governments historically emphasized tracking natural resources for the sake of posterity and avoiding overuse (Roosevelt), features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today.

While the importance of the environment as a factor for economic activity waned in the eyes of governments during the 20th century, it became increasingly salient as a source of waste management and a determinant of human health. The relationship between governments and the environment changed from one of planning for "wise use" to protection and regulation. Governments' use of environmental data shifted from tracking the productive potential of the local environment to monitoring indicators used for evaluating and enforcing policy. This paradigm shift in environmental measurement coincided with the development of modern national statistical systems that drove a massive expansion of economic data collection and curation. Despite massive

efforts by governments in collecting data on environmental indicators associated with human and animal health outcomes, environmental assets (e.g., fresh water or fertile topsoil) remain segregated from economic accounts.[1] While national accounts standards call for the inclusion of many forms of natural capital as non-financial capital (European Commission and Development), few countries collect and organize these data with the same fidelity as output or factor inputs. When they do, they are seldom used to measure TFP.

Despite the limited progress incorporating natural capital in national accounts, environmental factors have become increasingly salient for private economic agents. Insurers, producers, and households increasingly demand better environmental data (Yergin; Boomhower et al.). Unlike regulators, these agents track natural capital stocks because changes in environmental services have material consequences for their economic decision-making and create opportunities to profitably engage in conservation (Karolyi and Puente). We argue here that the persistent omission of environmental assets from national accounting frameworks also impedes government economic decision-making; the changing state of the environment is material for forecasting growth, irrespective of environmental or sustainability goals. Failing to account for how the environment determines economic growth may cause governments to overestimate the ability for society to substitute away from natural resources and misattribute the effects of economic policy (Nordhaus and Tobin; Brandt, Schreyer, and Zipperer). The results we present below suggest the potential magnitude of gains in measurement accuracy is be substantial. For the case of the median country in our sample (strangely enough, the United States), the estimated improvement in mean squared error for TFP growth over a 10-year horizon is 0.6 percentage point. This error reduction is roughly three times the annual year-on-year forecast error of the Congressional Budget Office's deficit projections, and corresponds to revenues larger than the estimated aggregate cost of compliance with the Clean Air Act (Congressional Budget Office; U.S. Environmental Protection Agency).

Existing Literature: Early work by Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (towards their broader measure of economic welfare) as well as the productive value of land (Nordhaus and Tobin). Subsequent work incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to reproducible capital (e.g., Mankiw, Romer, and Weil; Caselli; Caselli and Feyrer; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis). These studies apportion capital income between returns on physical investment and resource rents in order to correct for the effects of the latter when comparing the marginal product of produced capital and its share of national income across countries (Caselli and Feyrer; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis. Broadly, the literature has highlighted the importance of correcting for resource rents when calculating the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains high (Hsieh and Klenow). We add to this literature in characterizing explicitly how introducing previously-omitted factors

[1] For example, it was only in the early 2020s that the United States began developing economic accounts for land (Wentland et al.).

of production has ambiguous effects on the bias in measurement of of output elasticities when the additions are marginal and do not render the growth-accounting equation correctly specified.

A nascent strain of literature focuses explicitly on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to that of Brandt, Schreyer, and Zipperer, who were the first to our knowledge to study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuel. The authors show these adjustments depend on the relative rate of growth in the use of non-renewable resources to that of changes in other inputs, which leads measurement of natural capital to have ambiguous effects on growth estimates. Finally, Freeman, Inklaar, and Diewert highlight the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular constructions of TFP (e.g., those deployed in the Penn World Tables Feenstra, Inklaar, and Timmer). We add to this literature in explicitly characterizing under what conditions the measurement of non-renewable resources will reduce errors in estimating TFP as opposed to the magnitude of adjustments induced.

The paper proceeds in three steps. Section 1 tests whether existing, widely used metrics of TFP suffer from omitted variable bias due to the omission of natural capital stocks beyond those with readily available market prices. We show prima facie evidence that a modest subset of natural capital stocks are strongly correlated with benchmark measures of international productivity from the academic literature and national statistical agencies. Section 2 derives sharp conditions under which including a partial accounting of natural capital improves measurement of TFP. Even with perfect measurement of all inputs in production, the inclusion of previously-omitted natural capital stocks has ambiguous effects on the bias and error of estimates of productivity growth. Finally, Section 3 uses data on natural resources from the World Bank's Changing Wealth of Nations (CWON, World Bank) to calibrate an empirical analog of the theoretical setting in Section 2. Using estimated growth in indices measuring the aggregate use of non-renewable and renewable natural capital at the country level between 1996 and 2019, we show that for plausible parameters including non-renewable capital when estimating TFP reduces in-sample error across our entire sample of 99 countries with available data.

1 Residual-Based Productivity Measurement

Neoclassical economic theory attributes growth in output to two sources: factor inputs and innovation. The use of factor inputs (e.g., capital and labor) and resulting output are measured by national statistical agencies at regular frequencies. Residual changes in observed output, those unexplained by growth in inputs, are called the Solow residual and are imputed to productivity (Solow). This approach to measurement typically omits environmental inputs outside of those that directly provide market goods and services where factor income is observed in government accounts.

To characterize the theoretical implications of omitting natural capital when estimating productivity growth, consider a model economy where output at each instant $Y(t)$ depends on a vector of factor inputs— consisting of services from capital, labor, non-renewable resources, and renewable resources $\boldsymbol{X}(t) \triangleq \{K(t), L(t), N_1(t), N_2(t)\}$. The frontier governing what can be produced by various combination of these inputs is given by a production function F. In addition to factor inputs $\boldsymbol{X}(t)$, modeled final output is determined by a catch-all variable for total factor productivity:

$$Y(t) = TFP(t) \times F(\boldsymbol{X}(t)) \tag{1}$$

TFP encapsulates all other determinants of output beyond those inputs that are arguments in the production function at a given point in time. It is a broad measure of efficiency and the scope of productive technology available to society. TFP growth is a parsimonious modeling tool capturing how innovation and creative destruction expand the society-wide production possibility frontier holding resources used as inputs constant (Romer; Aghion and Howitt).

A critical aspect of the neoclassical production approach is that the the factor inputs in equation 1 are flows of *services* put towards production (e.g., the number of cycles by a turbine, total hours worked by the labor force, or the amount of crude oil or groundwater extracted). These flows are distinct from the stocks that create them; idle machinery, unemployed persons, and dormant subsurface minerals are not counted as inputs to production. However, in many cases measuring the totality of service flows entering production from a given resource is difficult. For example, the total harvest from a fishery in a given year may account for the harvest as an input in the fishing industry, but it will not capture the service provided by the remaining stock of fish whose reproduction helps replenish the fishery going into the next year (Hannesson). Total crude oil production will capture the input of extracted petroleum, but will fail to measure how a lower level of reserves reduces underground pressurization and increases future extraction costs (Anderson, Kellogg, and Salant). In cases like those above changes in stocks over time are imputed to changes in flows of services. Growth across all factor inputs is then calculated as the change in how all stocks of resources—natural, human, or produced—are used over time.

Standard assumptions for the functional form of the production function F (e.g., the Cobb-Douglas or transcendental logarithmic forms) yield a log-linear form with output growth equal to innovation (growth in TFP) plus a weighted sum of the growth in each element of a set of factor inputs $\boldsymbol{X}$:

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \boldsymbol{X}(t)} \gamma_x g_x(t) \tag{2}$$

where the $g_x(t)$ terms are the instantaneous growth rates of variables in $\boldsymbol{X}$ and γ_x are the output elasticities for produced capital, labor, and natural capital.[2] Equation (2) is known as the growth accounting equation.

[2]Output elasticities need not be fixed over time (e.g., if production takes a constant elasticity of substitution form) but we adopt that assumption here.

TFP is a theoretical aggregate index representing the horizon of productive technology available at a given point in time; it is by construction unobservable to the econometrician. In practice, TFP growth is typically measured as the portion of output growth not explained by growth in measured factor inputs, which we group together as the set $\boldsymbol{I}$ (Comin). Rearranging the growth accounting equation and swapping $\boldsymbol{X}$ for measured inputs $\boldsymbol{I}$ yields the empirical estimator for TFP, the Solow residual:

$$g_A \triangleq g_Y - \sum_{i \in \boldsymbol{I}} \gamma_i g_i \tag{3}$$

The residual between observed output growth as and growth across measured inputs indexed by $\boldsymbol{I}$ (Solow). When growth in output, all inputs, and their associated output elasticities are estimated without measurement error, $\boldsymbol{I} = \boldsymbol{X}$, growth in the Solow residual is an unbiased estimator of of TFP growth in (2). Versions of equation (3) are used by statistical agencies like the Bureau of Labor Statistics in the United States (Meyer and Harper) and the European Statistical Office in the European Union (European Commission and Eurostat) to construct measures of productivity growth in national accounts, and in academic applications for comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer). Estimates of historical TFP growth inform growth predictions that shape the revenue side of government budget forecasts (e.g., 10-year budget forecasts done by the Congressional Budget Office or the President of the United States Congressional Budget Office; Council of Economic Advisers) used to balance future outlays against receipts.

If a national accountant were able to calculate the dynamics of all existing natural capital stocks and their output elasticities in production, setting $\boldsymbol{I}$ to include all observed stocks would always improve her understanding of innovation vis-à-vis any other candidate for $\boldsymbol{I}$ (see Section 1 of the Supplemental Appendix). Therefore, a goal may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg). However, in the interim, limits on institutional capacity for the scope and precision of measurement will likely cause the set of measured inputs $\boldsymbol{I}$ to differ from $\boldsymbol{X}$ and cause some error when estimating the growth rates and output elasticities of factor inputs. Consider a case where a national accountant includes only two inputs, produced capital and labor, when estimating TFP (i.e., in our running example setting $\boldsymbol{I} = \{K, L\}$ and omitting both forms of natural capital). Conditional on her choice of inputs ($\boldsymbol{I}$) and her estimates of factor input growth rates and output elasticities ($\hat{\boldsymbol{g}}, \hat{\boldsymbol{\gamma}}$) she applies, the Solow residual-based estimator of TFP growth can be decomposed as

$$g_A | \boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x g_x}_{\text{Omitted natural capital inputs}} + \underbrace{\sum_{i \in K, L} [\gamma_i g_i - \hat{\gamma}_i \hat{g}_i]}_{\text{Measurement error}} \tag{4}$$

where hats denote estimated values. When natural capital inputs are omitted, (or in the general case whenever $\boldsymbol{X} \setminus \boldsymbol{I} \neq \emptyset$) (3) is a biased estimator for TFP growth; the contributions of the omitted

natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual. As a result, output growth driven by (potentially unsustainable) drawdowns of resources may be conflated with (sustainable) technological change. This misattribution can effect three negative outcomes for policymakers irrespective of environmental concerns: 1) an overestimation of the ability for society to substitute away from natural resources (Nordhaus and Tobin); 2) a failure to account for the depletion of natural resources required for replicating historical development paths (Brandt, Schreyer, and Zipperer); and 3) misallocation of resources (Hsieh and Klenow). In turn, a residual in (4) displaying substantial correlation with changes in a given natural capital stock over time is evidence that that natural asset is an omitted input in production. Conversely, a precise null of zero correlation between natural capital stocks and estimated TFP growth is suggestive that the empirical growth accounting equations are correctly specified and changes in natural capital are unlikely to bias current measures of growth. We now turn to describing how we test the null hypothesis of zero correlation between natural capital and existing measures of TFP in the data.

1.1 Hypothesis Testing

Existing measures of productivity in national accounts and research datasets are calculated using Solow residual-based methods that ignore natural capital (Meyer and Harper; Feenstra, Inklaar, and Timmer; European Commission and Eurostat). As shown in the decomposition in (4), these methods lead to three pathways for natural capital to be correlated with measured changes in TFP: (1) correlation between natural capital use and unobservable growth in TFP, (2) the omission of natural capital stocks which should enter (4), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While we cannot distinguish between these three potential mechanisms, we can test whether the Solow residual-based estimator of TFP is uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with TFP growth, we cannot rule out channel (2). Improved measurement of natural capital is then less likely to be immaterial for measuring TFP.

In an ideal world we would test channel (2)—whether N_i is an omitted variable—in isolation. However, as we only observe g_A jointly as the sum of the three terms in (4), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. This leads to two potential sources of type one errors where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i} but spuriously attribute it to N_i being omitted. The first two cases correspond to channels (1) and (3); when g_{N_i} is systemically correlated with true innovation, measurement error in the growth accounting equation, or both. Finally, there is a potential source of type two errors: if g_{N_i} is an omitted variable in the growth accounting equation and correlated with either measurement error or TFP growth, it is possible these correlations offset one another. The econometrician may then correctly fail to reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute this to natural capital playing no role in production.

Other theoretical barriers exacerbate the challenge of falsifying that estimates of g_A absorb the growth effects of changes in the use of natural capital. First, estimating output elasticities γ_x is nontrivial even for the case of physical capital where factor income can be attributed (Caselli and Feyrer); estimation is more difficult for labor and material inputs and at best a nascent practice for natural capital (Arrow et al.). In short, it is difficult to rule out that measurement error in factor input growth that enter existing estimates of TFP growth is non classical. Second, identifying the growth of natural capital inputs requires delineating what constitutes the flow of natural capital services that enter production from traditional measures of natural capital, such as stocks. For example, the amount of local cooling or precipitation provided by agriculture-adjacent forests may change substantially year-on-year despite de-minimus changes in canopy cover or biomass (Grosset, Papp, and Taylor). Finally, correlations between input growth and the theoretical object g_{TFP} are untestable, as the latter cannot be observed. While measures of some forms of innovation exist (e.g., patent filings and seed varietals Acemoglu et al.; Moscona and Sastry), we cannot rule out underlying correlations between environmental change and the broad measure of technological progress we seek to estimate. Our estimation strategy below then relies on three assumptions in order to be of use in testing whether natural capital inputs are omitted: (1), measurement error in the (weighted) factor input growth terms used to construct measured TFP is classical; (2), changes in natural capital stocks are a good proxy for changes in the flow of services they produce; and (3), the rate of true innovation g_{TFP} is conditionally uncorrelated with the changes in environmental inputs to production we consider.

1.2 Estimating Equation

To test the hypothesis of whether growth in natural capital stocks is associated with growth in measured TFP, we regress measured country-level TFP growth ($g_{A_{j,t}}$) from the Penn World Tables (Feenstra, Inklaar, and Timmer) on measured growth in seven country-level natural capital stocks ($g_{N_{n,j,t}}$) taken from the 2024 vintage of the World Bank CWON dataset. Under the second assumption described in Section 1.1 (that growth in natural capital stocks is directly proportional to the growth in the services that they provide) this allows us to identify the estimands γ_i. Equation (5) shows the empirical analog of (4) that can be taken to available data; we estimate (5) on a balanced panel of 117 countries over the 1996-2019 period:

$$g_{A_{j,t}} = \sum_{n \in \boldsymbol{N}} \gamma_n g_{N_{n,j,t}} + \sum_{z \in \boldsymbol{Z}} \theta_z g_{z,j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \tag{5}$$

where j and t index countries and years respectively. The set $\boldsymbol{N}$ contains growth in the seven natural capital inputs at the country level listed in Table 2 of the Supplemental Appendix and γ_i are the associated output elasticities for each input. The summation over terms $g_{z_{j,t}}$ allows for controls for growth in produced capital, employment, the labor share, and human capital at the country-by-year level taken directly from the PWT (Feenstra, Inklaar, and Timmer). Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ are idiosyncratic error

terms which we assume to be conditionally uncorrelated with growth in the natural capital stocks.

Table 1 shows estimated $\hat{\gamma}$ coefficients under various specifications of (5), each of which gives how changes in the growth of a given natural capital stock affects expected growth in TFP at the national level. To test whether natural capital stocks are omitted variables when accounting for output growth, our estimand of interest is the vector of output elasticities of the natural capital inputs $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_7]'$. The null hypothesis is then:

$$H_0 : \boldsymbol{\gamma} = \mathbf{0} \tag{6}$$

where $\mathbf{0}$ is the zero vector. Failing to reject H_0 suggests that the collection of all natural capital stocks in the CWON plays no role in the production function and omitting them is unlikely to bias current metrics of TFP. We interpret rejecting the null in (6) as suggestive evidence that natural capital assets enter the production function and bias Solow residual-based estimators of TFP when omitted. The bottom rows of table 1 show p-values from two Wald tests of this joint null hypothesis across all seven natural capital stocks we consider. The first row constructs test statistics using asymptotic standard errors (clustered at the country level), while the second uses a semi-parametric wild bootstrap approach (Wu). In specifications with time and country fixed effects (column 3 onward), we can reject that natural capital has no role in explaining output at traditional significance levels under both tests considered. In addition, almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs to production and covary positively with output.[3] While the estimated joint effects of changes in natural capital appear precise, the overall residual variation explained by adding natural capital on the margin is limited. Including natural capital increases model R^2 by 0.006 (3 percent over baseline levels) for the benchmark fixed effects models in columns (3) and (4).

Robustness: Partial correlations and individual tests of coefficient significance suggest that land rents are the natural capital inputs most likely to be material for current TFP metrics, aligning with suggestions for improving national accounting given by the authors of the latest edition of the PWT (Feenstra, Inklaar, and Timmer). However, even when we take the extreme measure of excluding urban, agriculture, and timber land from the set of natural capital in order to avoid any bias due to land use change affecting the measurement of capital income, the null hypothesis is still rejected (see Section 2.3 of the Supplemental Appendix). Additional tests verifying the robustness of the results for other subsets of independent variables (e.g., excluding natural capital stocks with a high share of missing or imputed values, expanding them to include non-renewable natural capital), choice of metric for TFP (e.g., measures from national statistical agencies and industry-specific industry based measures), and sample period can be found in Section 2.3 of the Supplemental Appendix.

[3]The negative coefficient on forest cover may be due to larger growth in forest cover reflecting a declining rate of forest land being converted to structures or agricultural land, which are likely positively associated with TFP growth.

Table 1: TFP Growth vs. Growth in Renewable Natural Capital

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.301	0.303	0.205	0.215	0.149	0.060
	(0.095)	(0.098)	(0.067)	(0.075)	(0.062)	(0.118)
Timber Land	0.164	0.166	0.071	0.075	-0.003	0.026
	(0.090)	(0.075)	(0.074)	(0.071)	(0.068)	(0.104)
Agricultural Land	0.027	0.043	0.030	0.036	0.041	0.043
	(0.025)	(0.025)	(0.022)	(0.023)	(0.021)	(0.017)
Forest Cover	-0.067	-0.111	-0.005	0.012	0.083	-0.027
	(0.094)	(0.091)	(0.079)	(0.083)	(0.053)	(0.076)
Mangroves	0.122	0.067	0.357	0.327	0.318	-0.108
	(0.244)	(0.242)	(0.299)	(0.322)	(0.367)	(0.427)
Biomass	0.112	0.026	0.094	0.072	0.047	-0.104
	(0.098)	(0.077)	(0.108)	(0.101)	(0.135)	(0.108)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.007	0.003	0.007	0.005	0.002	0.039
Bootstrap Wald	0.092	0.063	0.046	0.021	-	0.056
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

2 The Marginal Inclusion of Natural Capital

Section 1 gives evidence that the omission of natural capital biases existing measures of TFP. We now examine under what conditions including observed changes in natural capital improve estimated productivity growth. How much do these omissions matter for the precision of our existing estimates, and under what conditions will existing estimates be improved through including the limited set of natural capital stocks we measure today? While intuition suggests that including a previously-omitted determinant of growth should improve our estimates of TFP, it is not the case. Consider the residual based growth estimator in our running four-input example. Suppose an accountant measures growth in output, labor, produced capital, and non-renewables as well as the associated output elasticities, but cannot measure growth in society's use of renewable resources over time (g_{N_2}). If she elects to omit non-renewables when calculating TFP growth, substituting this into (4) gives that her estimand becomes

$$\hat{g}_A \triangleq g_{TFP} + \gamma_1 g_{N_1} + \gamma_2 g_{N_2} \tag{7}$$

If she instead includes non-renewables (g_{N_1}), the estimand for TFP becomes

$$\tilde{g}_A \triangleq g_{TFP} + \gamma_2 g_{N_2} \tag{8}$$

Clearly, it is ambiguous whether (8) is a less biased estimate of g_{TFP} than (7) without making an assumption on the sign of how the use of renewable inputs has changed over time. Section 1 of the Supplemental Appendix shows how the results generalize when multiple inputs may be simultaneously omitted; bias is reduced only when all omitted terms share a sign in expectation. Criteria for improving in-sample metrics of goodness of fit (such as root mean squared error) are even more stringent, as they depend on higher moments for growth in the omitted stocks and the set of included inputs. The effects of including natural capital on the margin, even with perfect measurement of growth and estimation of output elasticities, are only guaranteed to improve estimates of TFP when it leads the estimator to be correctly specified (i.e., $\boldsymbol{X} \subseteq \boldsymbol{I}$).

The result that marginal inclusion of natural capital stocks need not improve estimates of TFP growth (or output elasticities) is our central addition to this literature. For instance, existing work by Brandt, Schreyer, and Zipperer shows that under the assumption that non-renewable capital inputs are uncorrelated with the use of any other natural capital stocks, TFP growth inclusive of non-renewable natural capital tends to be higher than when it is measured excluding non-renewable natural capital. However, because of the results shown here, it is not possible to know whether these reconstructed series which adjust for non-renewable resources actually bring use closer to estimating true productivity growth over time.

3 Calibrated Estimates of Incorporating Natural Capital on the Margin

Section 2 showed that the effects of marginal additions of inputs to the growth accounting equation, even when perfectly measured, are not guaranteed to improve residual-based TFP estimates. The final section here examines when the addition will improve estimation conditional on the behavior of observed growth in natural capital we see in the data. Here, we examine theoretical conditions under which measuring the growth rate of a natural capital stock used in production and including it on the margin improves estimation of TFP growth. Our laboratory returns to the model economy governed a four-factor production function. Like in the example of Section 2, suppose a national accountant seeks to measure TFP growth in this economy, but measures only growth in the usage of produced capital, labor and non-renewable capital, N_1. Growth in all of these quantities (along with output) are measured without error. While she suspects it is important in production, she has no ability to measure the use of renewable resources (N_2) over time. Knowing this, she must choose whether to include growth in societal use of non-renewable resources, N_1, when taking Equation (4) to the data. Her criteria for improved TFP measurement is expected in-sample mean squared error for her data which cover 70 years.

To evaluate how the decision to include natural capital may affect the resulting measurement error in her estimates of TFP, we conduct a thought experiment where each estimator of applied TFP growth is applied to simulated processes for output growth calibrated to historical values across a broad set of countries. To operationalize this process, we follow the World Bank CWON methodology (World Bank) and construct aggregate quantity indices for non-renewable and renewable resource use at the country level and their growth, $g_{N_{1,j,t}}$ and $g_{N_{2,j,t}}$, over time (see Section 3 of the Supplemental Appendix for details). We estimate empirical averages and correlations for how the use of non-renewable capital, renewable capital, and produced capital change over time in 99 countries between 1972 and 2019. We pair these constructed values of growth in annual natural capital use with measured use of produced capital, labor, and TFP taken from the Penn World Tables (Feenstra, Inklaar, and Timmer), a benchmark publicly-available dataset with harmonized measures of input growth across countries.

After calibrating data generating processes for each country, we derive how introducing non-renewable resources N_1 in the growth accounting equation affects the expected root mean squared error (RMSE) in residual-based estimates of TFP growth when output elasticities are estimated using ordinary least squares.[4] This, in effect, allows us to compare the accuracy of estimates generated by (7) against (8). We evaluate the expected RMSE at the sample analogs of population moments input growth under standard assumptions for the output elasticities of various factors of production. This gives us a cross section of 99 countries for the effects of including natural capital when calculating TFP growth, which can be conditioned on observed levels and correlations

[4] See Section 4.3 of the Supplemental Appendix for the full procedure for applying (7) and (8) to our simulated data.

between the use of natural capital over time at the national level.

3.1 Results

Figure 1 plots the results from our calibrated thought experiment. Each cell overlain with an ISO3 code corresponds to the pair of empirical correlations between the growth of non-renewable capital inputs and produced capital inputs (x axis) and the growth of non-renewable and renewable capital inputs (y axis) we observe in the data for that country between 1996 and 2019. For each cell with an ISO code, the outcome is the change in the expected RMSE for estimated TFP growth when non-renewable capital is included on the margin when taking versions of the estimators (7) and (8) to the data:[5]

$$\Delta\text{RMSE}|\boldsymbol{g} \triangleq \sqrt{\mathbb{E}[(\hat{g}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{g}]} - \sqrt{\mathbb{E}[(\tilde{g}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{g}]} \tag{9}$$

A positive change in our RMSE metric for a given hypothetical data generating process for growth in factor inputs and TFP $\boldsymbol{g}$ indicates that including non-renewable resources improves a national accountant's estimate. The z axis in Figure 1 plots these conditional changes in expected RMSE— $\Delta\text{RMSE}|\boldsymbol{g}$— for each country in our calibrated cross section; darker blue indicate smaller improvements while lighter greens show progressively larger ones. The remaining values within the convex hull formed by our cross section (dashed gray lines) are correlation pairs which do not correspond to the historical behavior of a specific country and are fitted using linear interpolation. The improvement surface then posits the improvement for a hypothetical country when a national accounting includes non-renewable capital when calculating TFP conditional on historical correlations in the growth of capital inputs over time.

The most striking result in Figure 1 is that for all 99 countries we consider, our simulations indicate that including non-renewable resources improves the accuracy of estimated TFP growth regardless of its correlation with the omitted renewable capital input. While the minimum RMSE reduction is quite close to zero (Tunesia, where we find $\Delta\text{RMSE} = 4.23 \times 10^{-8}$ percent), the median (mean) value is 6 (19) basis points. The improvement is disperse (s.d. 40 basis points) and right skewed. The magnitude of this improvement is typically about 10% of the magnitude of average TFP growth, for which the median values are 67 (43) basis points for countries where it is positive (negative) on average for the 1996-2019 period.

Figure 2 shows a set of scatter plots that show the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of TFP growth. Panels 2a and 2b plot improvements in estimates of TFP growth against various average growth in g_{N_1} and g_{N_2} at the country level. 86 countries (79%) exhibit positive average growth in their use of non-renewables and 58 (53%) experience positive growth in renewables usage. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process (set as the

[5]The full derivation of this process is relegated to Section 4 in the Supplemental Appendix.

observed value in the PWT). Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. The median (mean) improvement is 12 (22) percent of baseline error when both natural capital stocks are omitted, and rises to 33% for over a quarter of the countries in our cross section of hypothetical outcomes.

The resulting improvement for most countries is of a magnitude that is policy-relevant. Median reductions in RMSE are 6 basis points basis points (1/100th of a percentage point). An RMSE of predicted growth of 6 basis points will amount to a forecast error of roughly 0.6 percentage point of TFP growth over a 10-year horizon. For the case of the United States (which for the RMSE reduction is the median country), this is three times the annual CBO year-on-year forecast error when predicting budget deficits (Congressional Budget Office). The budget implications of this forecast error compounded over ten years leads to differences in revenue roughly equal in size as the aggregate public and private cost of compliance with the Clean Air Act in the United States between 1970 and 1990 in magnitude, one of the most broad environmental policies enacted globally to date (U.S. Environmental Protection Agency).

Comparison to existing results in the literature: Our estimates of the mean squared error reduction in estimates of annual TFP are similar to those produced by Brandt, Schreyer, and Zipperer when they consider adjusting existing metrics for a subset of non-renewable resources we consider here. The authors find find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries they consider between 1986 and 2008. Freeman, Inklaar, and Diewert use an earlier version of the Changing Wealth of Nations dataset to calculate the bias improvements when only subsurface minerals and fossil fuels matter for output growth and are added to the growth accounting equation on the margin. Their findings suggest that omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies such as those in OPEC or those with large mining sectors. Our estimates for the reduction in estimation error are about two orders of magnitude smaller than the year-on-year revisions of TFP estimates from the United States' Bureau of Labor Statistics to their estimates for 2024 (U.S. Bureau of Labor Statistics).

4 Conclusion

This paper presents evidence for how measuring natural capital can aid governments irrespective of environmental or public health goals. When natural capital affects production, omitting it from national accounts biases measures of economic growth. This supports the original assertion by Nordhaus and Tobin in 1972 that leaving things out of national accounts "gives the wrong signals for the directions of economic growth." (Nordhaus and Tobin). Existing literature continues to highlight how environmental change affects society beyond health outcomes : sparrow eradication campaigns lead to lower yields and higher mortality 1950s China, and declines in bat populations caused lower land rents in the United States (Frank et al.; Manning and Ando). Even for governments that do

not see the health or preservation aspects of environmental outcomes a priority, natural capital has first-order consequences for measuring and forecasting economic growth. The reductions in error from improved measurement will influence objectives such as deficit reduction and growth targets that are almost always concerns for governments irrespective of their position on the intrinsic value of the environment to society.

The size of existing errors, when measuring TFP, are economically meaningful even for horizons as short as ten years. Historical growth in inputs and measured productivity between 1996 and 2019 suggests over half of countries globally may improve their TFP estimates of TFP by 10 percent or more by including non-renewable resources when estimating productivity growth. Even for countries where this improvement is small, measurement of natural capital in the interim will still help with retrospective calculations of TFP as the full set of inputs governing the economy becomes clearer over time. In the case of the United States, for an "America's Balance Sheet" policy to improve the quality of economic measurement, it is important to have a broad set of natural capital measures and not simply a narrow focus on non-renewable resources and surrounding industry (U.S. Department of the Interior). Comprehensive measurement will bolster federal ability to track productivity growth and correctly price natural and produced assets, accounting for the interdependence between both in production.

Finally, the posterity value of measurement undertaken today may be high even for resources that lack a role in current production processes. Expanding the set of measured stocks is substantially easier when done contemporaneously (c.f. when done retrospectively). Fenichel argues a critical feature of natural capital is understanding a small number of accounting boundaries, in part to separate resources currently entering production from those that are not. Yet, innovation itself, measured by TFP, can change what is important for innovation. Having measurements in hand can help address this "new inputs" challenge in measuring the economy (Trajtenberg). As future generations develop new productive technologies that draw on previously-unused environmental services, improved measurement will enable future generations to achieve commensurate progress in informing future fiscal forecasts.

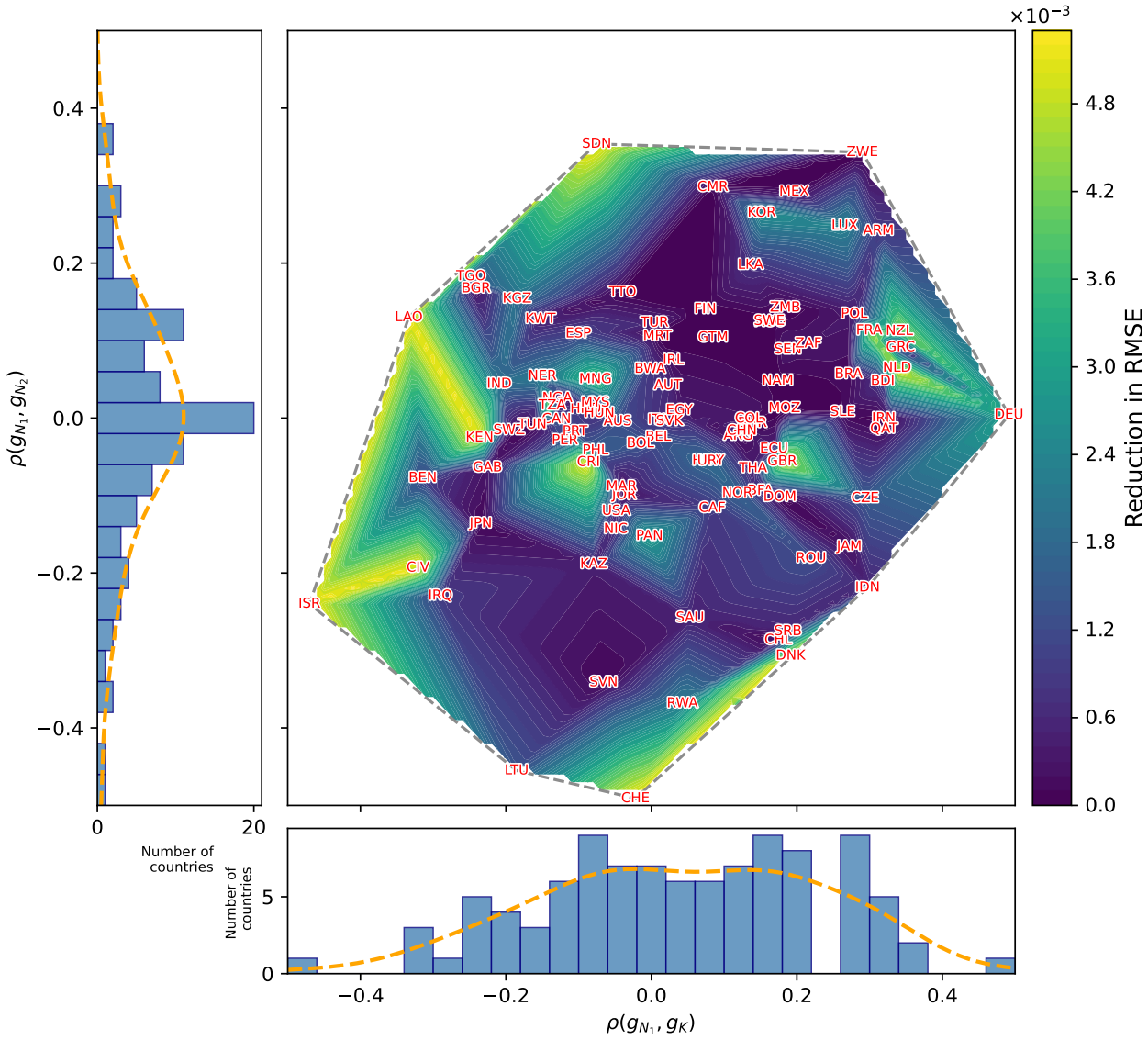

Figure 1: Contour plot shows the reduction in the root mean squared error (RMSE) when adding g_{N_1} to the residual-based TFP estimator. RMSE reduction are plotted against observed correlations between factor inputs $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ at the country level between 19956 and 2019 for the 99 countries we consider. Contour lines denote level sets of the RMSE reduction. The dashed polygon formed by the gray lines indicates the convex hull formed by the set of countries where sufficient data are available. Values of the z variable plotted within the convex hull are fitted using piecewise linear interpolation and top coded at the 90th percentile (Δ RMSE $= 0.006$). Histograms on each axis show the empirical distributions of $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ across the countries in our sample.

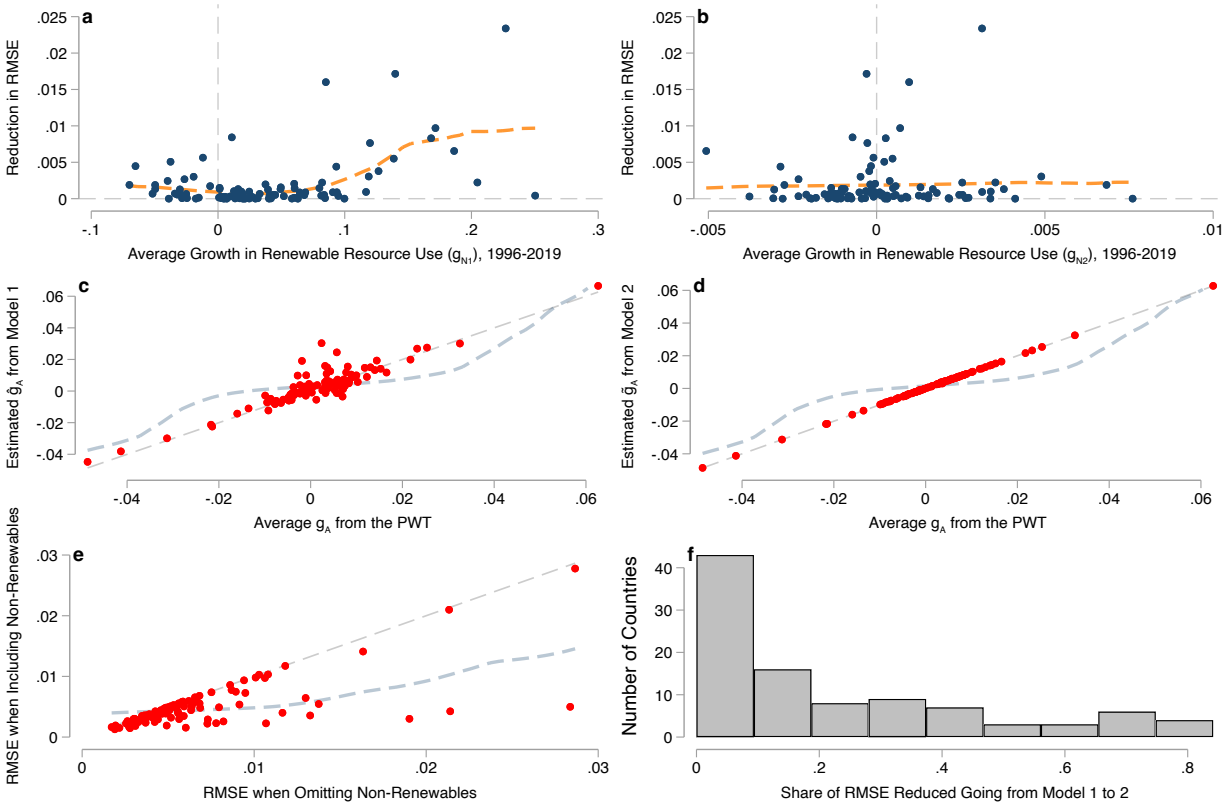

Figure 2: Simulated outcomes plotted against various sample moments. Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 99 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines shows fitted values from a local linear regression using the Epanechnikov kernel with a rule-of-thumb bandwidth (Fan).

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